EVOLUTIONARY COMPUTATION

LABORATORY 01

# Greedy heuristics

for the Traveling Salesperson Problem with Node Costs

ASSIGNMENT 1 - REPORT

SOURCE REPOSITORY

https://github.com/Flat-Earther/Evolutionary-Computation/tree/main/lab-01-greedy-heuristics


Authors

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OBJECTIVE

## Minimize cycle length + selected node costs

Choose exactly ceil(n / 2) nodes and arrange them into a closed cycle.

The supplied instances contain 200 nodes, so every solution selects 100.

INSTANCES

### TSPA and TSPB

200 nodes each - 100 selected

METHODS

### 4 greedy and baseline methods

200 solutions evaluated per method

# 1. Problem description

The input consists of n nodes, each with integer planar coordinates (x, y) and an integer node cost.

The task is to choose exactly k = ceil(n / 2) distinct nodes and specify their cyclic visiting order.

The cycle must return from its last selected node to its first selected node.

OBJECTIVE

**f(S) = cycle length(S) + sum of costs of selected nodes**

DISTANCE

d(i,j) = round(sqrt((xi-xj)^2 + (yi-yj)^2))

Distances are calculated once
and stored in a distance matrix.

DATA SEPARATION

Optimization methods access
the distance matrix and node
costs, not the original coordinates.

**For the supplied TSPA and TSPB instances, n = 200 and k = 100.**

This problem combines a subset-selection decision with a cyclic routing decision. The algorithms below construct feasible solutions directly and evaluate every candidate change using both distance and node cost.

## Solution representation

A solution is an ordered list of k distinct node indices.

The closing edge from the last node to the first is included in the objective.

All reported indices are zero-based, matching the supplied solution checker.

# 2. Algorithms and pseudocode

## 2.1 Random solution

Randomly selects the required number of distinct nodes. It is used as a baseline.

```
Algorithm RandomSolution(n, k)
1. V <- {0, 1, ..., n - 1}
2. S <- randomly select k distinct nodes from V
3. Return S

The cycle is closed when F(S) is evaluated.
```

## 2.2 Nearest Neighbor - End

Starts with one node and repeatedly appends the most promising unused node to the path end.

```
Algorithm NearestNeighborEnd(D, C, start)
1. V <- {0, 1, ..., n - 1}
2. S <- [start]; U <- V \ {start}
3. k <- ceil(n / 2)
4. While |S| < k:
 best <- argmin over v in U:
 D[last][v] + C[v]
 append best to S; remove best from U
5. Return S
```

## 2.3 Nearest Neighbor - Any Position

Maintains an open path and tests insertion at its beginning, end, and between adjacent nodes.

```
Algorithm NearestNeighborAnyPosition(D, C, start)
1. S <- [start]; U <- V \ {start}
2. k <- ceil(n / 2)
3. While |S| < k:
 For each v in U:
 test beginning: D[v][S[0]] + C[v]
 test end: D[S[-1]][v] + C[v]
 test each gap (a,b):
 D[a][v] + D[v][b] - D[a][b] + C[v]
 insert the best node at the best position
 remove the node from U
4. Return S
```

## 2.4 Greedy Cycle

Maintains a closed cycle and inserts the best unused node into the best cycle edge.

```
Algorithm GreedyCycle(D, C, start)
1. S <- [start]; U <- V \ {start}
2. k <- ceil(n / 2)
3. second <- argmin over v in U:
 2 * D[start][v] + C[v]
4. S <- [start, second]; remove second from U
5. While |S| < k:
 for each v in U and closed edge (a,b):
 delta <- D[a][v] + D[v][b]
 - D[a][b] + C[v]
 insert the best node and remove it from U
6. Return S
```

# 3. Computational experiment - TSPA

The experimental procedure consists of generating 200 solutions for each algorithm and instance.

For the greedy methods, solutions are constructed from different starting nodes; Random uses independent samples.

For each method, minimum, maximum, average, and standard deviation are calculated.

## Experimental results - TSPA

| Algorithm | Minimum | Maximum | Average | Std. dev. |
|---|---|---|---|---|
| Random | 233130 | 289417 | 260294.40 | 9730.46 |
| Nearest Neighbor - End | 84362 | 90075 | 87476.63 | 1328.22 |
| Nearest Neighbor - Any Position | **72328** | 77022 | 74346.97 | 981.62 |
| Greedy Cycle | 72590 | 75281 | **73793.41** | 527.45 |

lowest minimum  lowest average

## Result summary

The best individual solution for TSPA was found by Nearest Neighbor - Any Position, with objective 72328.

The best average result for this instance was obtained by Greedy Cycle, with average objective 73793.41.

**All solutions were checked with the provided solution checker.**

## Runtime reference

Random: 3 ms | Nearest Neighbor - End: 84 ms

Nearest Neighbor - Any Position: 7174 ms | Greedy Cycle: 7596 ms

# 3. Computational experiment - TSPB



The experimental procedure consists of generating 200 solutions for each algorithm and instance.

For the greedy methods, solutions are constructed from different starting nodes; Random uses independent samples.

For each method, minimum, maximum, average, and standard deviation are calculated.

## Experimental results - TSPB

| Algorithm | Minimum | Maximum | Average | Std. dev. |
|---|---|---|---|---|
| Random | 184618 | 235381 | 206384.53 | 8859.90 |
| Nearest Neighbor - End | 56007 | 63166 | **59397.34** | 2093.12 |
| Nearest Neighbor - Any Position | **52915** | 66164 | 59480.28 | 2171.96 |
| Greedy Cycle | 55655 | 63734 | 59649.91 | 2804.92 |

lowest minimum  lowest average

## Result summary

The best individual solution for TSPB was found by Nearest Neighbor - Any Position, with objective 52915.

The best average result for this instance was obtained by Nearest Neighbor - End, with average objective 59397.34.

**All solutions were checked with the provided solution checker.**

## Runtime reference

Random: 2 ms | Nearest Neighbor - End: 82 ms

Nearest Neighbor - Any Position: 7306 ms | Greedy Cycle: 7633 ms

**The following pages present the best solutions and their node-index lists.**

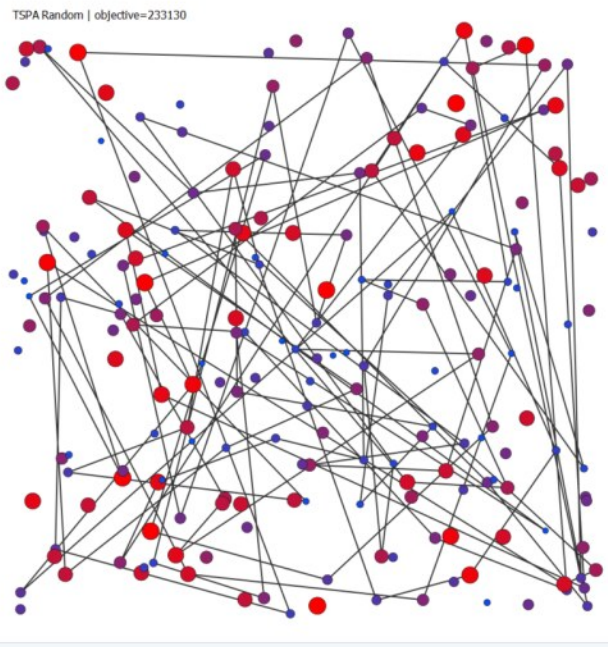
# 4. Best solutions - TSPA

The plots show node cost using colour and size. The line represents the closed cycle.

## Random

min=233130 max=289417 avg=260294.

**best = 233130**



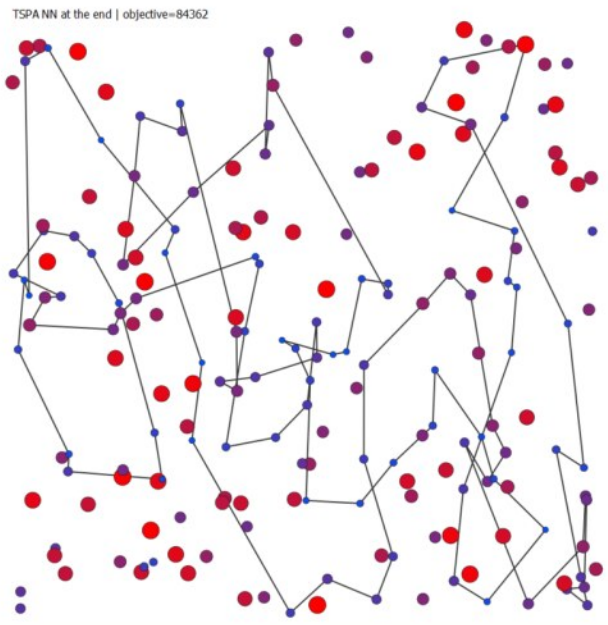

### Best solution nodes:

153 24 79 88 138 35 148 186 184 144 81 166 27 83
49 69 114 126 17 131 20 134 91 169 0 61 150 76
143 52 5 2 25 156 107 66 16 18 145 127 135 82 125
63 139 198 181 43 154 11 183 111 93 39 189 100
101 68 191 129 164 130 105 192 51 26 122 167 110
87 99 14 23 193 155 136 92 40 45 84 62 4 7 13 103
8 33 3 197 15 159 1 78 42 196 102 176 58 54 55
153

## Nearest Neighbor - End

min=84362 max=90075 avg=87476.63

**best = 84362**



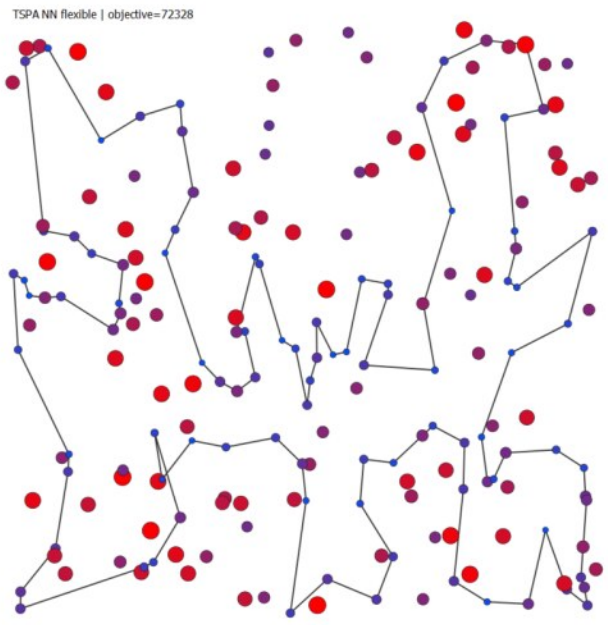

### Best solution nodes:

54 9 33 36 177 67 144 53 135 58 134 176 174 106
89 107 57 182 151 93 50 3 1 164 178 29 136 186
184 47 159 130 153 78 88 193 170 146 75 64 150 42
113 43 85 142 8 131 84 158 192 181 114 141 180
132 76 80 48 31 63 149 168 97 127 187 30 123 152
108 60 116 49 11 119 160 83 90 51 16 35 87 45 140
52 179 185 28 102 156 157 71 91 38 100 4 112 139
115 5 54

## Nearest Neighbor - Any Position

min=72328 max=77022 avg=74346.97

**best = 72328**



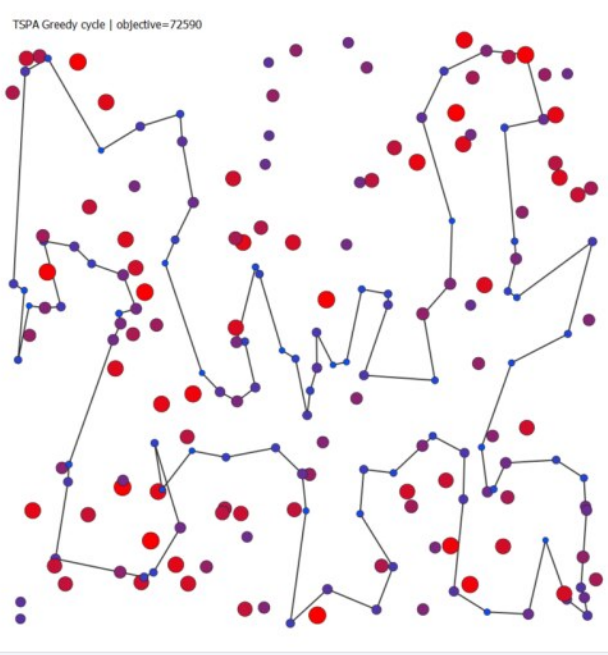

### Best solution nodes:

104 129 69 127 97 168 116 149 63 11 49 160 83 123
100 152 108 60 31 48 80 91 157 71 4 76 132 180 28
102 185 35 156 51 16 67 144 140 53 179 52 177 36
33 9 54 131 174 8 164 186 136 163 29 101 178 1
169 3 50 147 47 93 106 42 89 113 130 159 146 170
153 193 78 88 107 64 57 182 151 150 176 72 134 84
58 158 192 181 114 135 73 45 87 141 187 30 189 2
171 104

## Greedy Cycle

min=72590 max=75281 avg=73793.41

**best = 72590**




### Best solution nodes:

48 80 91 157 71 4 76 132 180 28 102 185 35 156 51
16 67 144 140 53 179 52 177 36 33 9 54 131 174 8
142 164 186 136 163 29 101 178 1 169 3 50 147 47
93 106 42 89 113 130 159 146 170 153 193 78 88
107 64 57 182 151 150 176 72 134 84 58 158 192
181 114 135 73 45 87 141 187 30 189 2 171 69 127
97 160 83 123 90 100 152 108 60 49 11 63 168 149
116 31 48

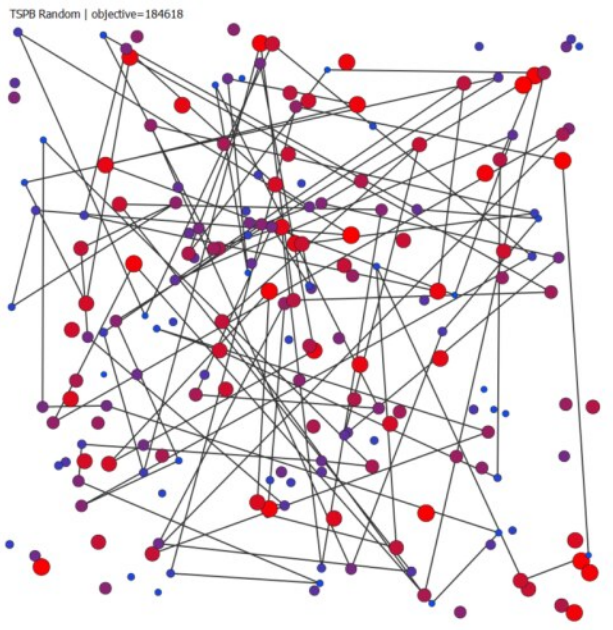
# 4. Best solutions - TSPB

The plots show node cost using colour and size. The line represents the closed cycle.

## Random

min=184618 max=235381 avg=206384.

**best = 184618**



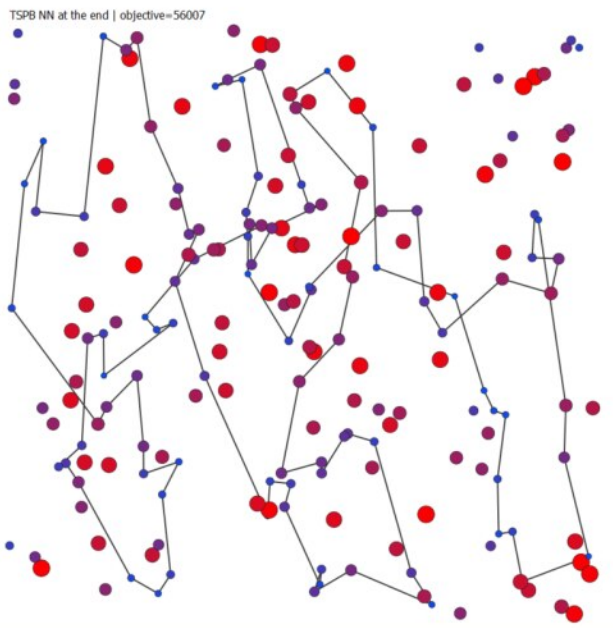

### Best solution nodes:

```
92 46 8 125 71 16 173 143 7 55 28 70 26 40 36 10
164 154 37 0 145 65 191 138 177 51 107 128 113 22
48 176 89 169 39 124 77 34 148 86 165 18 153 4
140 52 142 61 103 161 171 20 115 178 149 31 152 6
197 87 45 156 193 110 2 76 163 85 122 157 99 15
185 179 95 189 116 104 114 192 88 50 196 137 3
117 33 66 121 69 198 83 123 54 187 38 194 159 158
147 92
```

## Nearest Neighbor - End

min=56007 max=63166 avg=59397.34

**best = 56007**



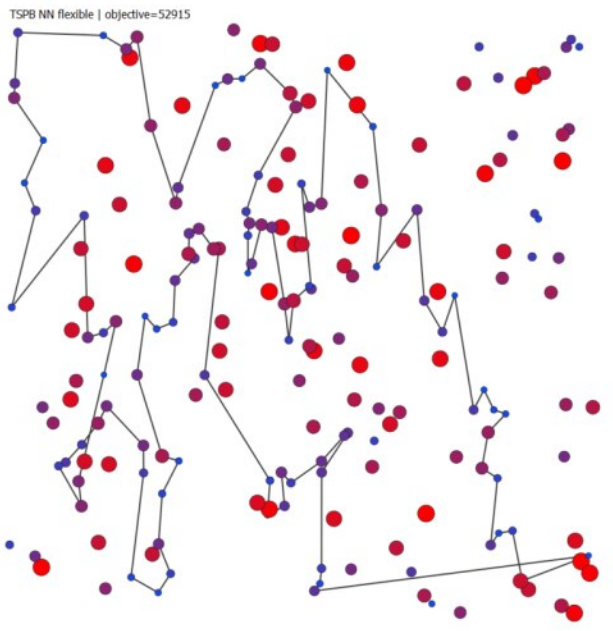

### Best solution nodes:

```
89 57 76 54 127 129 90 153 20 14 2 110 118 56 114
196 10 148 143 78 192 162 104 185 184 3 151 69
172 133 120 96 121 198 48 72 46 86 117 160 136 8
21 154 140 47 170 125 177 6 132 29 197 175 134 45
60 40 71 155 97 115 180 65 28 149 145 158 116 147
42 50 130 30 183 38 64 85 59 167 51 87 179 82 88
171 99 67 24 34 181 152 73 98 27 135 75 79 156 43
89
```

## Nearest Neighbor - Any Position

min=52915 max=66164 avg=59480.28

**best = 52915**



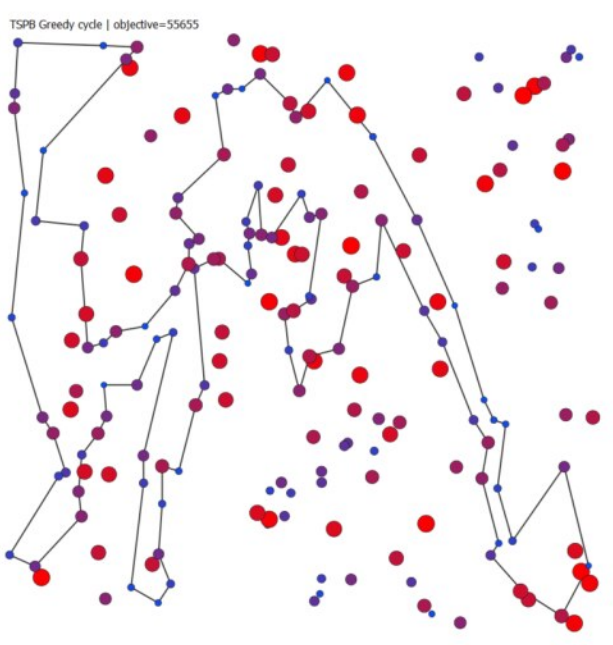

### Best solution nodes:

```
110 2 14 20 74 153 182 52 90 129 127 80 54 192
162 104 76 185 57 89 86 46 72 3 184 173 151 122
117 8 160 69 172 136 133 120 156 43 48 96 198 121
130 191 50 42 147 116 37 146 13 149 28 145 65 158
177 125 26 170 197 36 132 29 6 180 115 155 71 175
134 45 157 60 40 141 97 154 140 47 38 21 176 30
183 189 168 64 59 85 51 73 167 152 34 24 181 179
87 82 110
```

## Greedy Cycle

min=55655 max=63734 avg=59649.91

**best = 55655**




### Best solution nodes:

```
174 36 197 6 180 115 170 97 140 47 155 71 175 134
45 157 60 141 40 33 64 21 189 168 69 160 172 136
133 120 8 117 72 46 86 184 3 173 122 151 98 111
27 135 76 185 162 192 80 52 182 20 74 2 93 110
118 14 153 90 129 127 54 104 57 89 156 43 48 96
198 121 63 130 191 183 30 176 38 154 26 125 177
138 158 145 149 147 42 116 37 146 13 28 65 178
103 29 132 1 174
```

# 5. Conclusions

The experimental results show that the greedy algorithms outperform the Random Solution method.

Randomly selecting nodes produces inefficient routes with many long and crossing connections.

## Main conclusions

Nearest Neighbor - Any Position achieved the best individual solution on both instances: 72328 for TSPA and 52915 for TSPB.

The best average result depends on the instance. For TSPA, Greedy Cycle obtained the lowest average objective (73793.41). For TSPB, Nearest Neighbor - End obtained the lowest average objective (59397.34).

The visualizations show that Random Solution produces routes with many long and crossing edges, while the greedy methods generally produce more compact cycles by considering distance and node costs during construction.

Allowing insertion at different positions can improve the best solution found, although it does not always guarantee the best average performance. The greedy methods make irreversible local decisions and therefore do not guarantee the global optimum.

The obtained solutions provide a useful starting point for more advanced methods, such as local search, which could further improve route quality.

SOURCE REPOSITORY

**https://github.com/Flat-Earther/Evolutionary-Computation/tree/main/lab-01-greedy-heuristics**

**The worst solutions are presented in the final section as an additional comparison.**

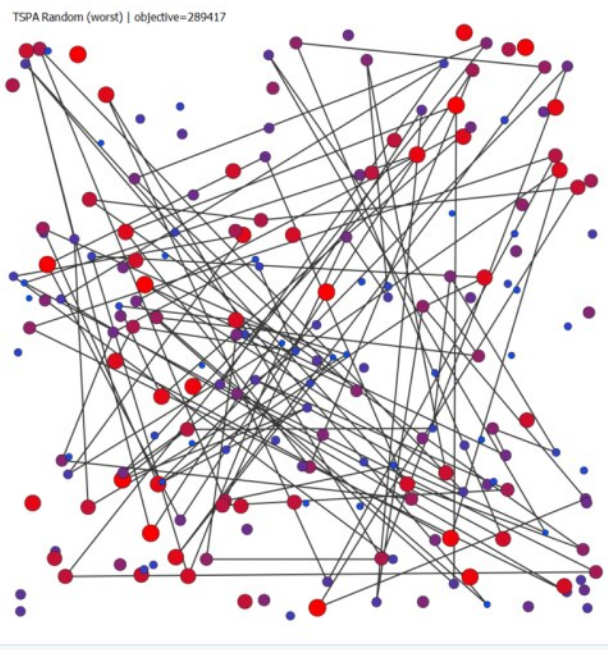
# 6. Worst solutions - TSPA

The plots show node cost using colour and size. The line represents the closed cycle.

## Random

min=233130 max=289417 avg=260294.

**worst = 289417**



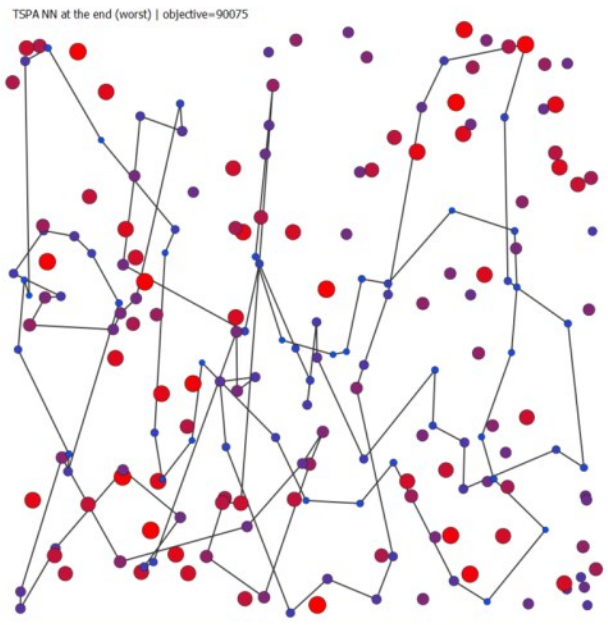

**Worst solution nodes:**

```
109 161 155 165 12 148 144 51 160 95 65 139 167
189 24 61 14 195 182 90 184 23 123 172 36 120 108
199 198 186 192 194 187 140 125 145 176 142 130
43 113 68 0 154 54 143 96 103 185 151 7 27 134
107 115 81 162 57 31 13 75 66 133 6 118 146 8 128
152 183 67 127 173 116 158 10 156 99 38 163 110
26 28 136 49 88 119 46 181 137 20 55 40 82 177
166 17 175 124 149 109
```

## Nearest Neighbor - End

min=84362 max=90075 avg=87476.63

**worst = 90075**



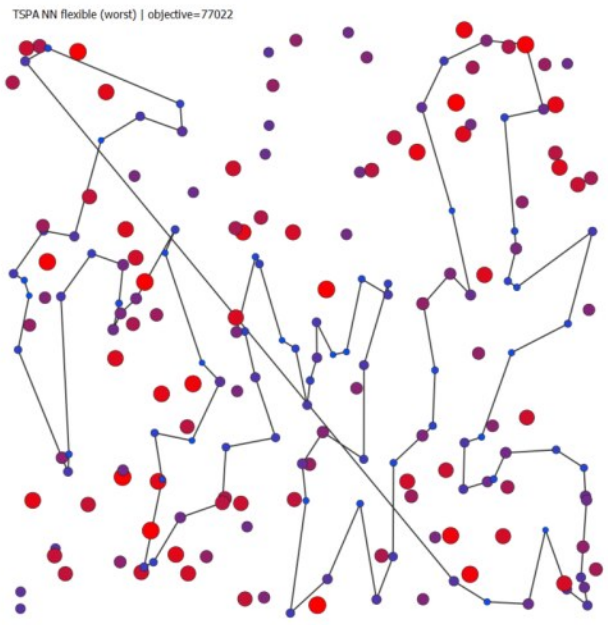

**Worst solution nodes:**

```
5 51 67 177 36 33 9 164 1 50 93 106 89 107 57 182
134 58 135 45 180 141 187 30 132 76 80 48 31 63
149 168 97 127 123 152 108 60 116 49 11 119 160
83 90 157 71 91 38 100 35 16 144 53 140 52 179 84
174 176 150 151 130 159 47 3 178 29 136 186 54
131 99 158 192 181 114 87 28 185 102 156 2 171
189 161 69 104 129 95 92 98 73 41 61 25 10 27 96
70 5
```

## Nearest Neighbor - Any Position

min=72328 max=77022 avg=74346.97

**worst = 77022**



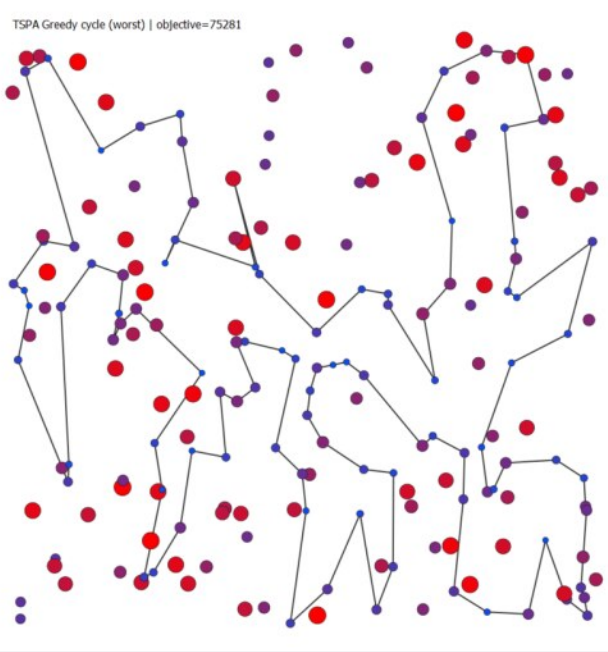

**Worst solution nodes:**

```
31 48 157 71 91 80 108 60 116 149 63 168 95 127
97 49 152 100 123 160 83 90 76 132 180 28 141 30
187 171 2 189 96 87 45 185 35 156 51 16 67 144
140 53 179 52 177 36 33 54 9 131 84 41 73 135 114
181 58 192 158 134 72 176 174 8 142 85 164 186
136 163 29 101 178 1 169 3 50 147 47 93 106 150
151 42 89 113 130 159 146 170 153 193 78 88 107
64 57 182 31
```

## Greedy Cycle

min=72590 max=75281 avg=73793.41

**worst = 75281**




**Worst solution nodes:**

```
18 16 52 33 9 54 174 8 142 164 186 136 163 29 101
178 1 169 3 50 147 47 93 106 42 89 113 130 159
146 170 153 193 78 88 107 64 57 182 151 150 176
72 131 36 177 179 53 140 41 84 134 158 192 58 181
114 135 73 45 144 67 35 156 185 102 28 87 141 189
2 171 187 30 180 90 83 160 123 100 152 49 97 127
168 63 149 116 60 108 31 48 80 91 157 71 4 132 76
51 18
```

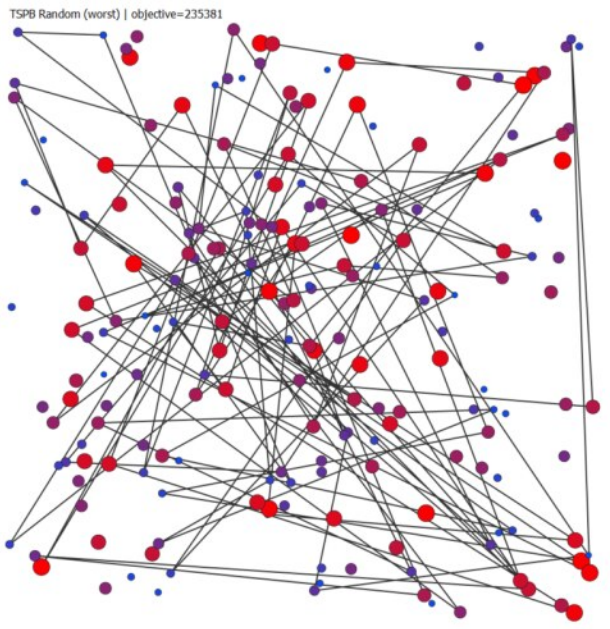
# 6. Worst solutions - TSPB

The plots show node cost using colour and size. The line represents the closed cycle.

## Random

min=184618 max=235381 avg=206384.

**worst = 235381**



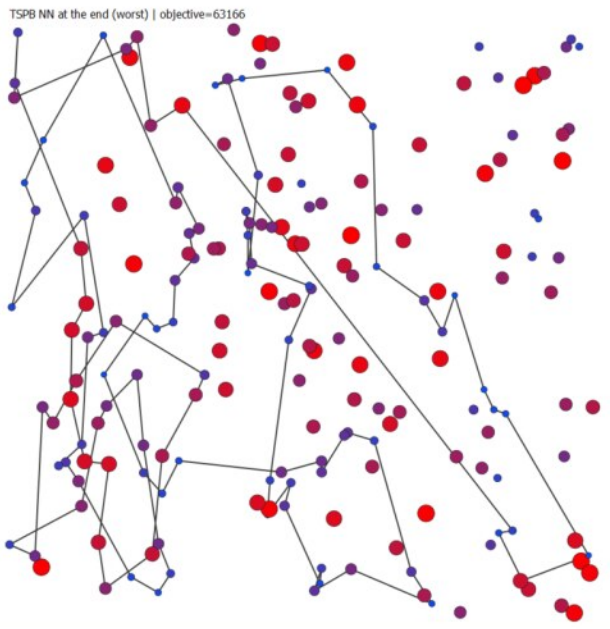

**Worst solution nodes:**

```
26 13 173 63 41 121 183 162 45 144 72 29 73 169
44 64 131 94 112 125 12 123 9 174 39 117 37 116
167 161 11 158 34 22 130 4 32 103 77 25 30 133 59
157 129 180 99 195 138 146 79 78 75 154 18 1 38 2
128 83 52 60 100 28 68 85 109 33 43 71 58 110 82
24 168 108 166 54 141 14 124 185 188 95 49 47 111
199 164 113 23 55 139 53 17 176 67 74 86 10 26
```

## Nearest Neighbor - End

min=56007 max=63166 avg=59397.34

**worst = 63166**



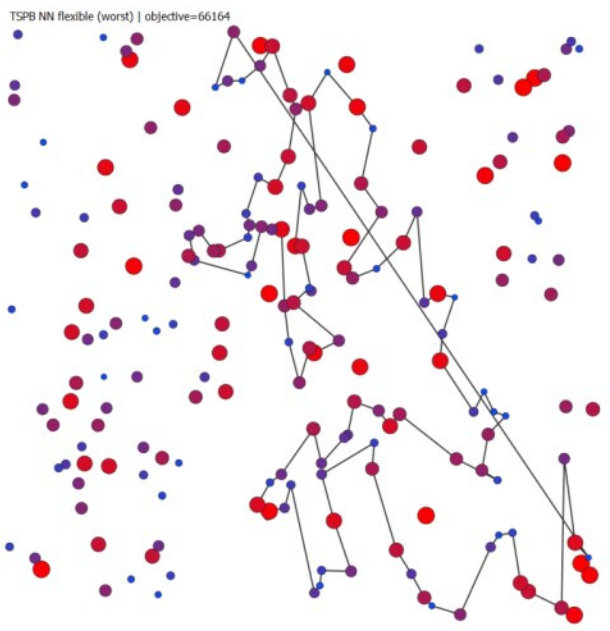

**Worst solution nodes:**

```
14 20 2 110 90 129 127 54 192 162 76 57 89 96 121
198 120 69 172 133 136 160 184 3 151 59 85 167 51
87 179 82 88 171 99 67 24 34 181 152 73 40 60 71
170 154 140 47 38 21 30 183 130 191 116 149 28
145 65 158 125 177 6 132 29 197 175 134 45 157
155 97 115 180 36 1 174 178 103 19 26 64 33 141
142 62 91 113 5 102 53 55 138 146 37 13 147 42 50
195 14
```

## Nearest Neighbor - Any Position

min=52915 max=66164 avg=59480.28

**worst = 66164**



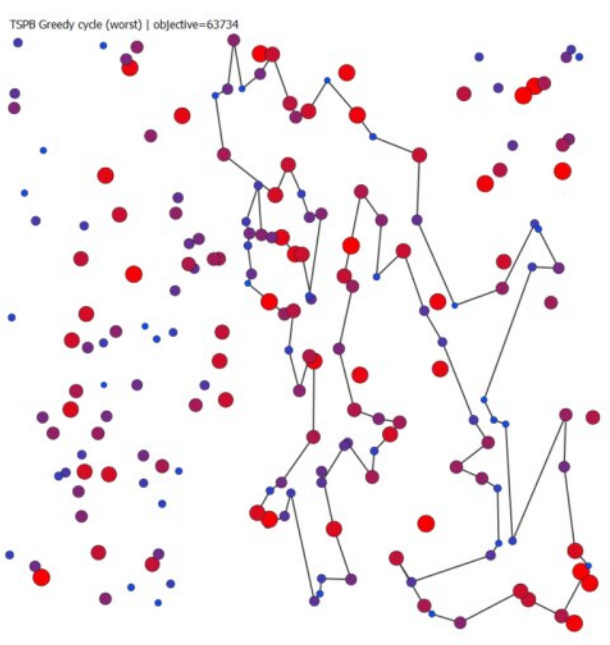

**Worst solution nodes:**

```
110 68 118 93 9 2 188 14 20 74 17 171 39 99 163
31 67 181 126 88 179 87 82 167 51 101 85 107 59
73 25 152 34 24 11 66 119 84 150 153 182 52 90
129 127 80 106 192 54 159 162 104 144 76 135 75
185 79 57 83 89 32 86 46 72 109 137 3 184 173 27
22 111 98 151 122 95 117 8 160 69 21 30 183 189
168 172 136 133 120 92 166 156 43 164 48 96 198
121 58 110
```

## Greedy Cycle

min=55655 max=63734 avg=59649.91

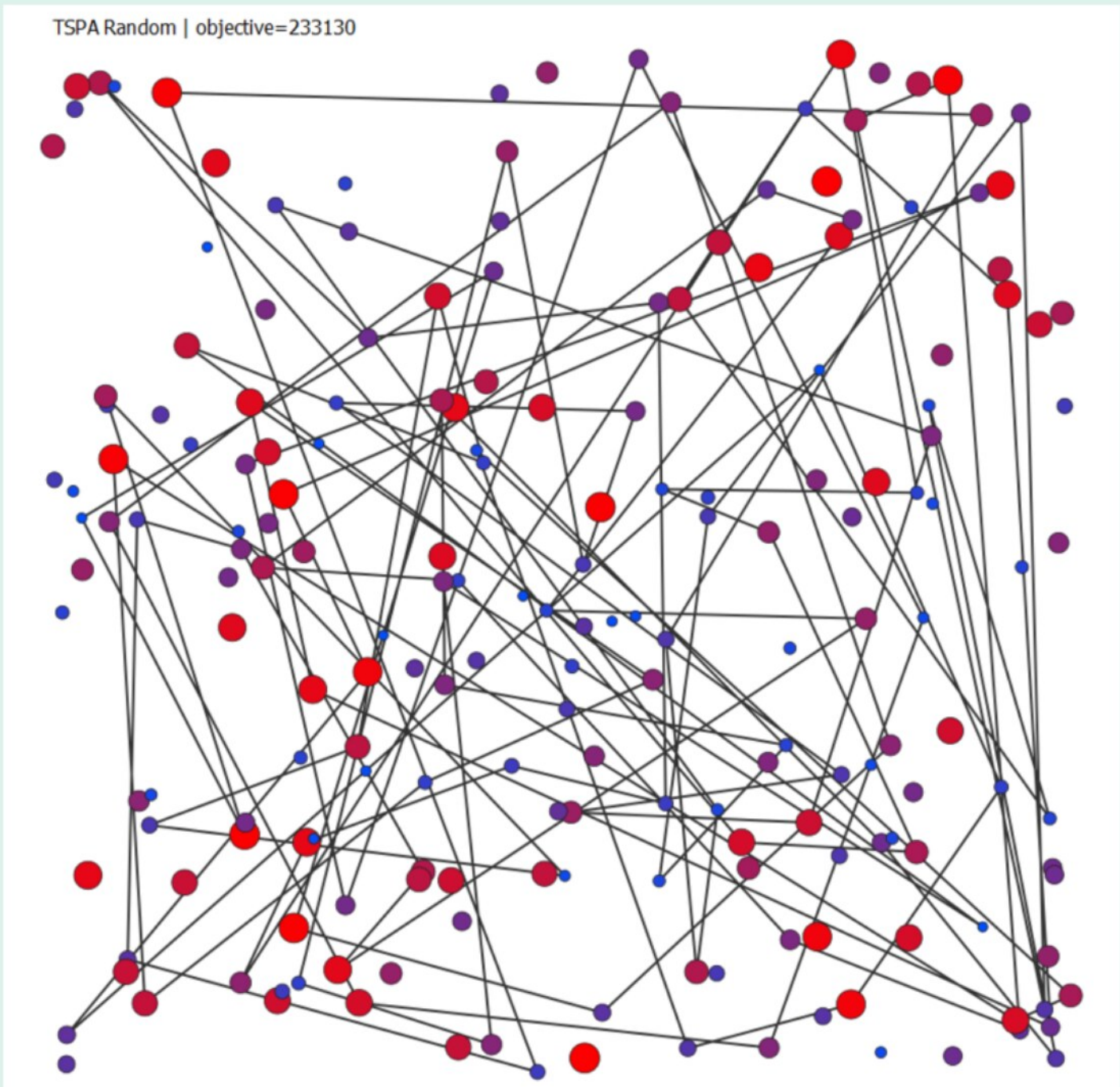
**worst = 63734**




**Worst solution nodes:**

```
163 39 171 17 188 2 9 93 110 68 118 56 14 90 129
127 10 148 114 196 78 54 104 77 57 83 89 32 156
43 164 48 96 58 198 121 63 92 166 72 46 86 184 3
137 109 95 117 8 120 133 136 172 160 69 122 173
151 98 111 22 25 73 59 107 85 51 167 82 87 179 88
126 152 181 34 24 31 67 119 84 66 11 27 135 75 79
185 76 144 162 192 80 52 150 182 153 20 74 99 163
```

# 7. Large best visualizations - TSPA




BEST SOLUTION

## Random

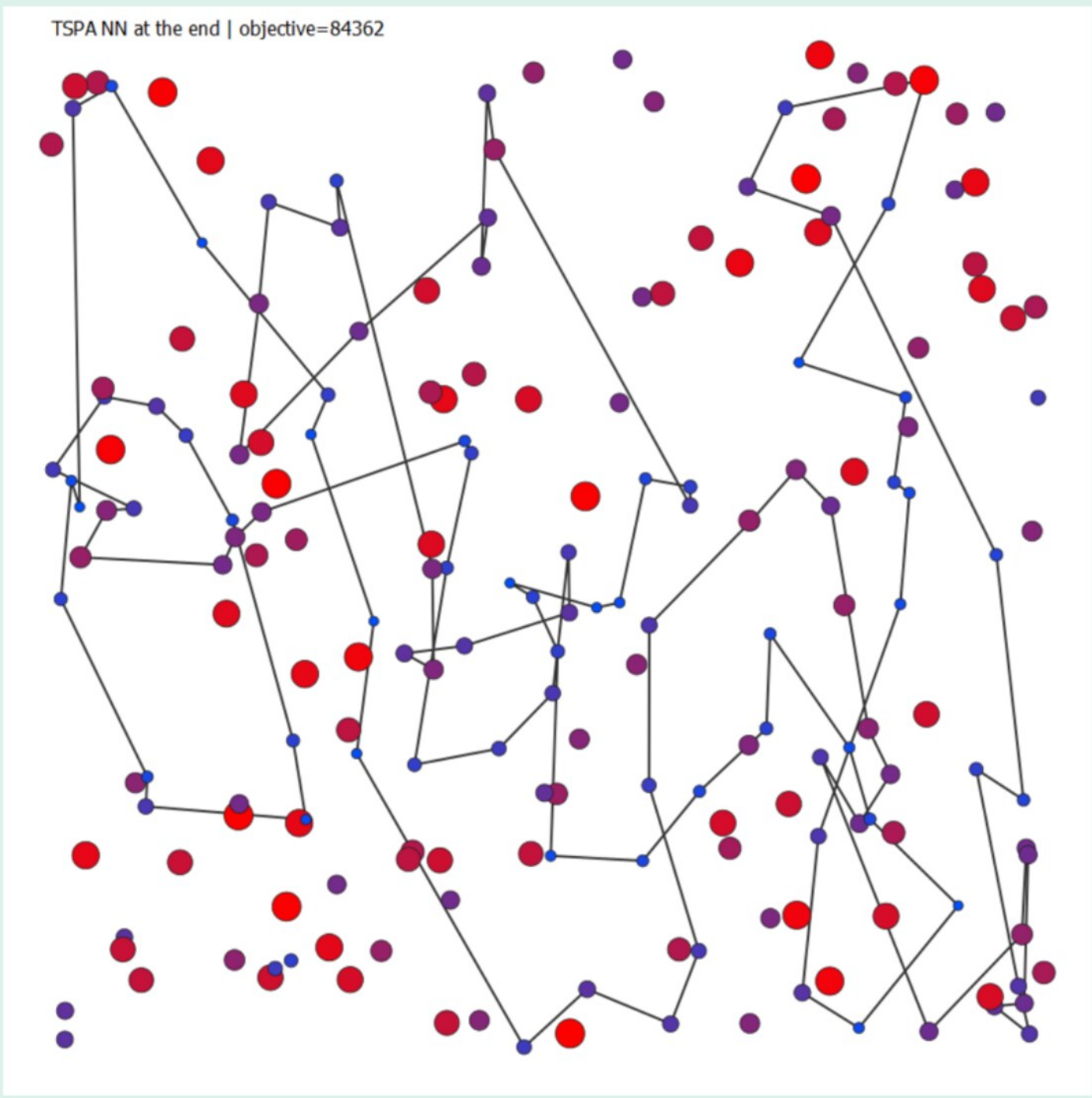
minimum 233130
maximum 289417
average 260294.40

objective 233130

### Best solution nodes

153 24 79 88 138 35 148 186 184 144 81
166 27 83 49 69 114 126 17 131 20 134
91 169 0 61 150 76 143 52 5 2 25 156
107 66 16 18 145 127 135 82 125 63 139
198 181 43 154 11 183 111 93 39 189
100 101 68 191 129 164 130 105 192 51
26 122 167 110 87 99 14 23 193 155 136
92 40 45 84 62 4 7 13 103 8 33 3 197
15 159 1 78 42 196 102 176 58 54 55
153

100 selected nodes; closed route




BEST SOLUTION

## Nearest Neighbor - End

minimum 84362
maximum 90075
average 87476.63

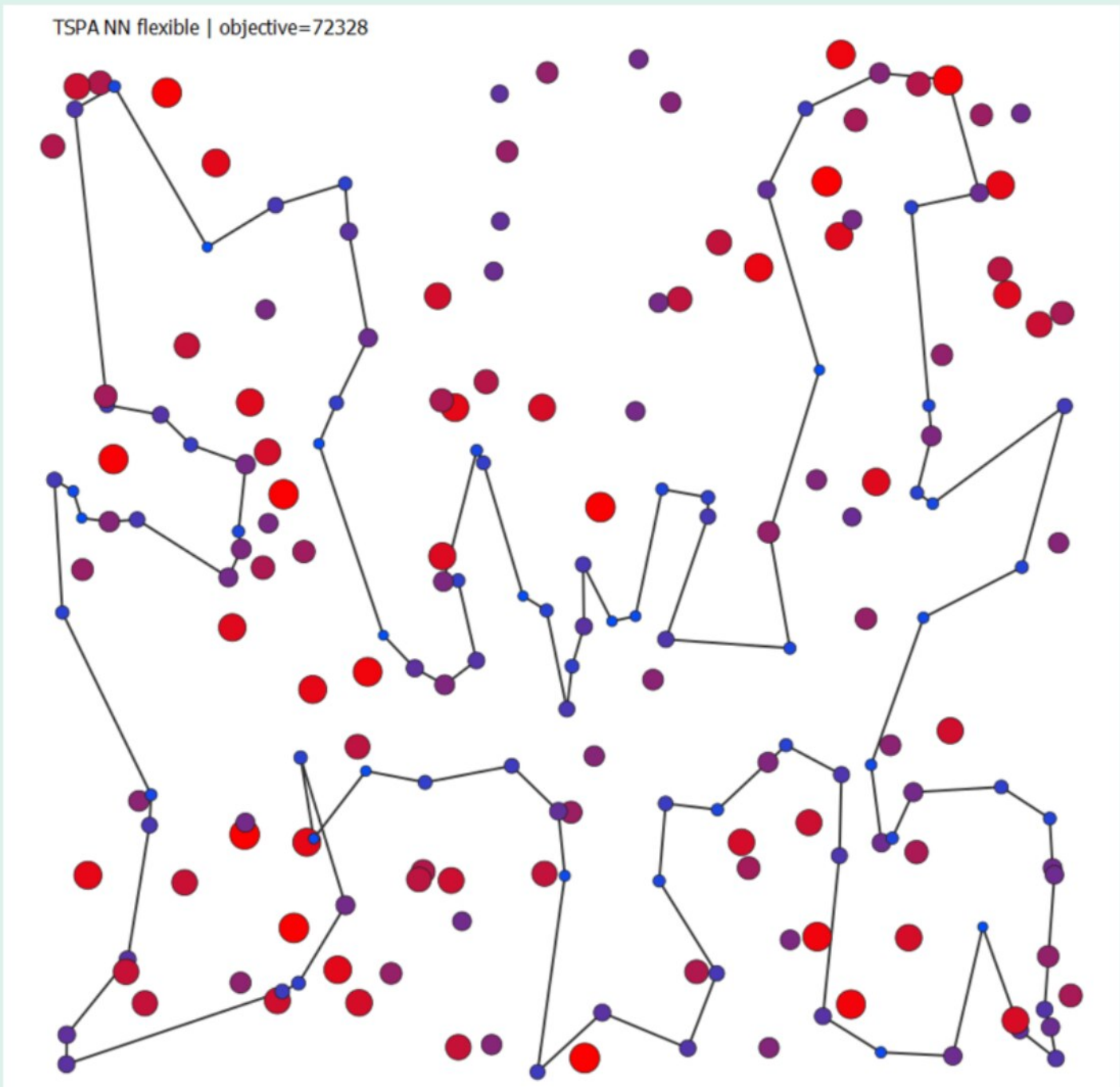
objective 84362

### Best solution nodes

54 9 33 36 177 67 144 53 135 58 134
176 174 106 89 107 57 182 151 93 50 3
1 164 178 29 136 186 184 47 159 130
153 78 88 193 170 146 75 64 150 42 113
43 85 142 8 131 84 158 192 181 114 141
180 132 76 80 48 31 63 149 168 97 127
187 30 123 152 108 60 116 49 11 119
160 83 90 51 16 35 87 45 140 52 179
185 28 102 156 157 71 91 38 100 4 112
139 115 5 54

100 selected nodes; closed route

# 7. Large best visualizations - TSPA




BEST SOLUTION

## Nearest Neighbor - Any Position

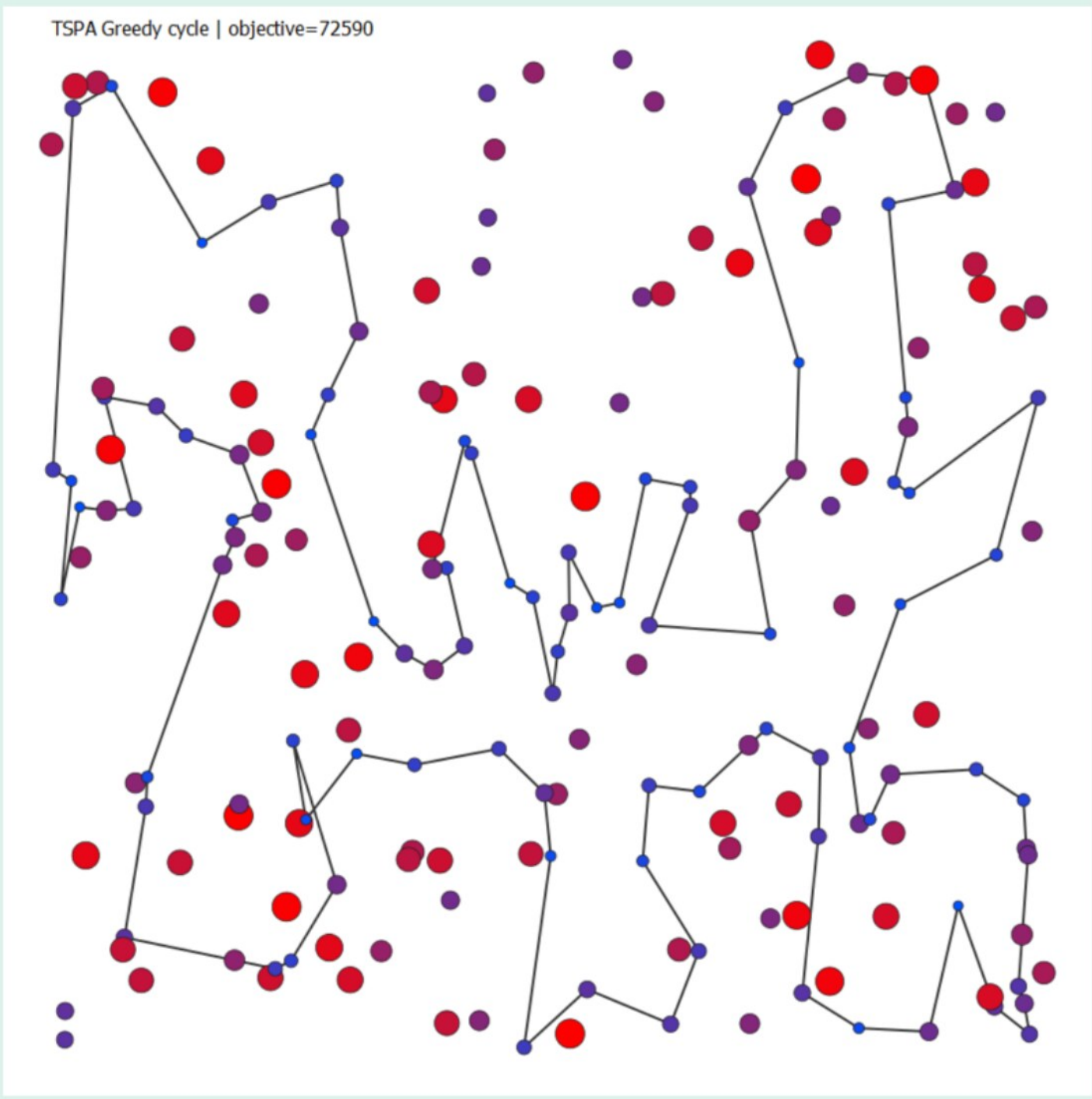
minimum 72328
maximum 77022
average 74346.97

objective 72328

### Best solution nodes

104 129 69 127 97 168 116 149 63 11 49
160 83 123 100 152 108 60 31 48 80 91
157 71 4 76 132 180 28 102 185 35 156
51 16 67 144 140 53 179 52 177 36 33 9
54 131 174 8 164 186 136 163 29 101
178 1 169 3 50 147 47 93 106 42 89 113
130 159 146 170 153 193 78 88 107 64
57 182 151 150 176 72 134 84 58 158
192 181 114 135 73 45 87 141 187 30
189 2 171 104

100 selected nodes; closed route




BEST SOLUTION

## Greedy Cycle

minimum 72590
maximum 75281
average 73793.41

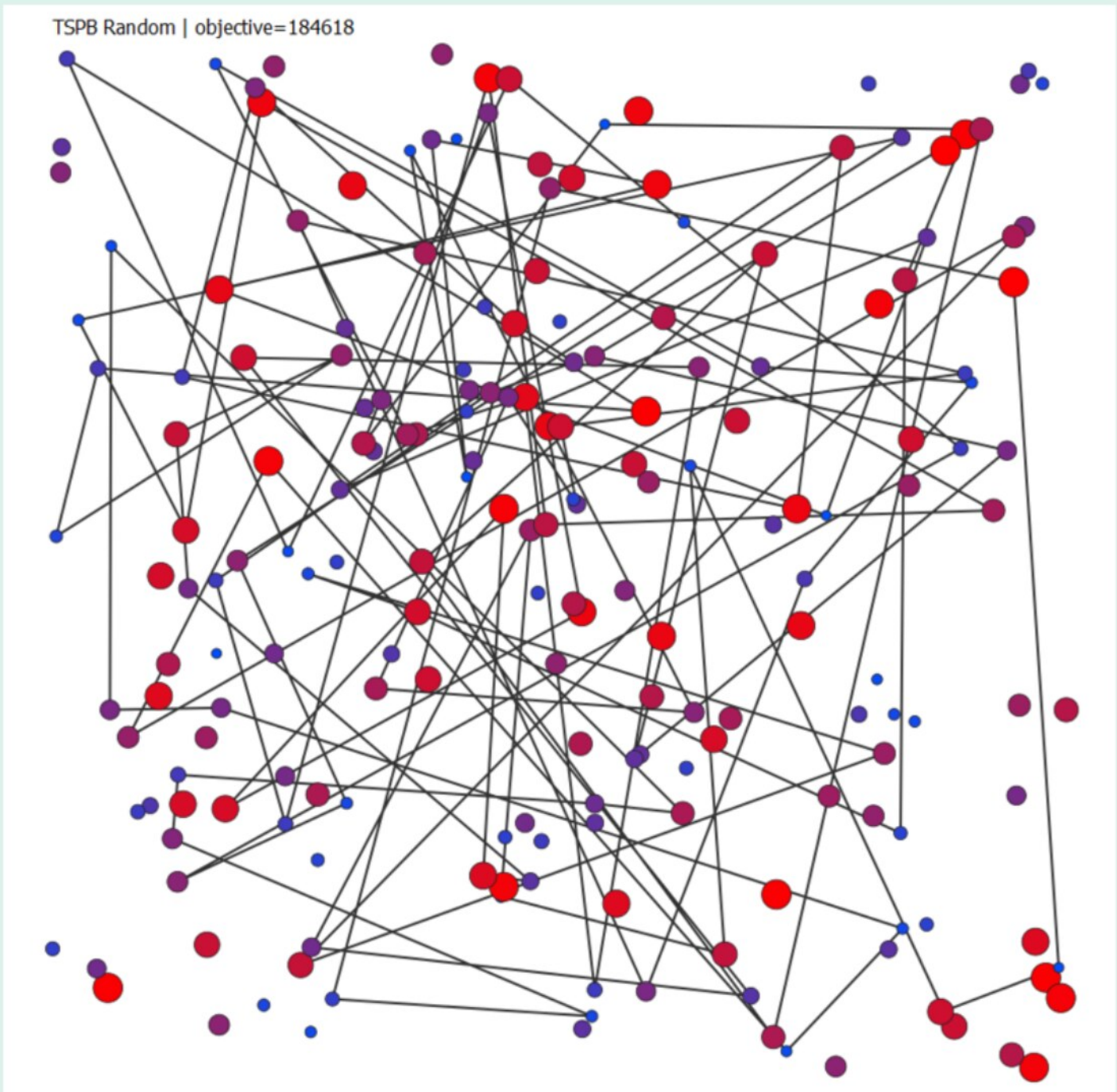
objective 72590

### Best solution nodes

48 80 91 157 71 4 76 132 180 28 102
185 35 156 51 16 67 144 140 53 179 52
177 36 33 9 54 131 174 8 142 164 186
136 163 29 101 178 1 169 3 50 147 47
93 106 42 89 113 130 159 146 170 153
193 78 88 107 64 57 182 151 150 176 72
134 84 58 158 192 181 114 135 73 45 87
141 187 30 189 2 171 69 127 97 160 83
123 90 100 152 108 60 49 11 63 168 149
116 31 48

100 selected nodes; closed route

# 7. Large best visualizations - TSPB




BEST SOLUTION

## Random

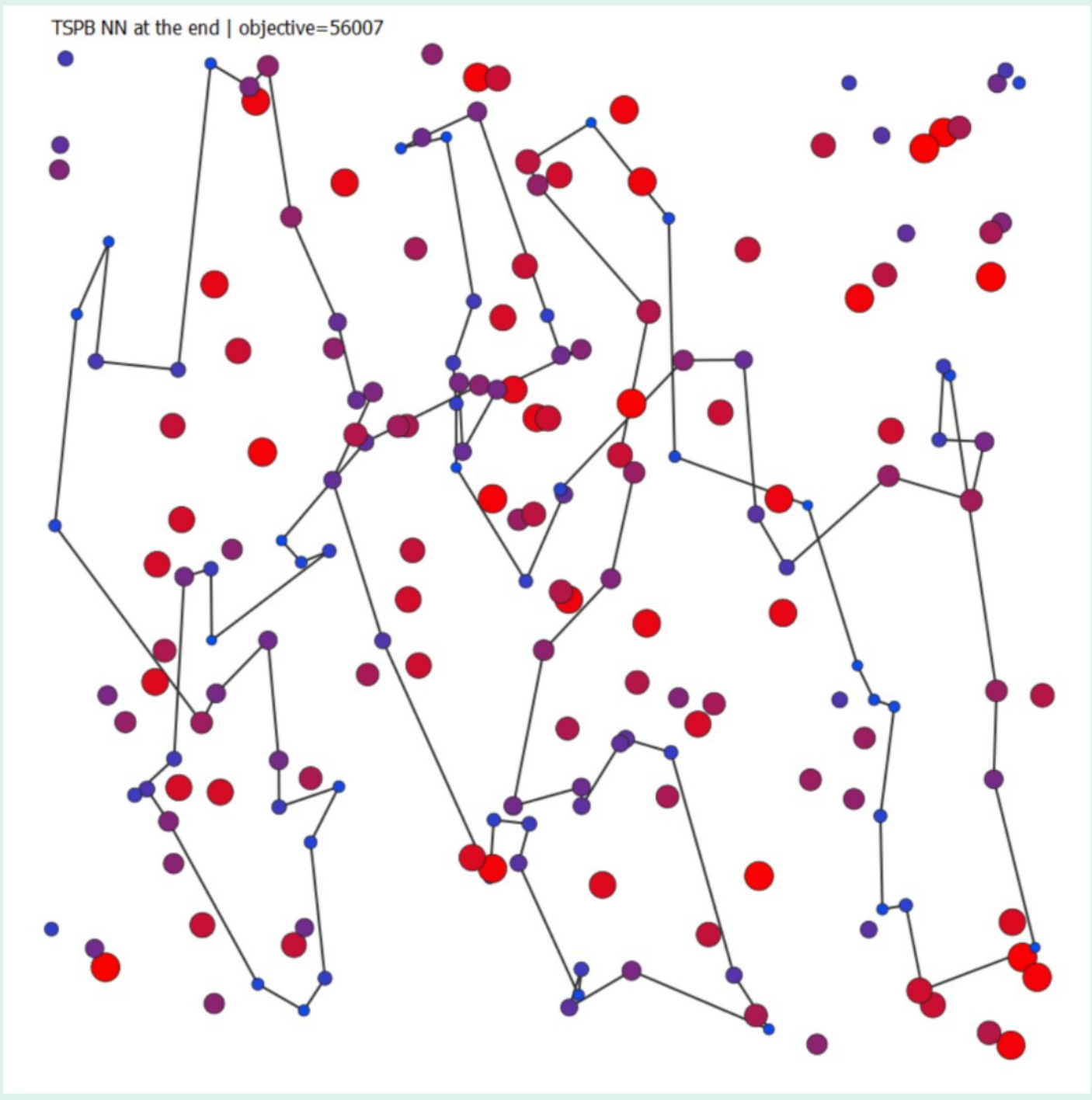
minimum 184618
maximum 235381
average 206384.53

objective 184618

### Best solution nodes

```
92 46 8 125 71 16 173 143 7 55 28 70
26 40 36 10 164 154 37 0 145 65 191
138 177 51 107 128 113 22 48 176 89
169 39 124 77 34 148 86 165 18 153 4
140 52 142 61 103 161 171 20 115 178
149 31 152 6 197 87 45 156 193 110 2
76 163 85 122 157 99 15 185 179 95 189
116 104 114 192 88 50 196 137 3 117 33
66 121 69 198 83 123 54 187 38 194 159
158 147 92
```

100 selected nodes; closed route




BEST SOLUTION

## Nearest Neighbor - End

minimum 56007
maximum 63166
average 59397.34

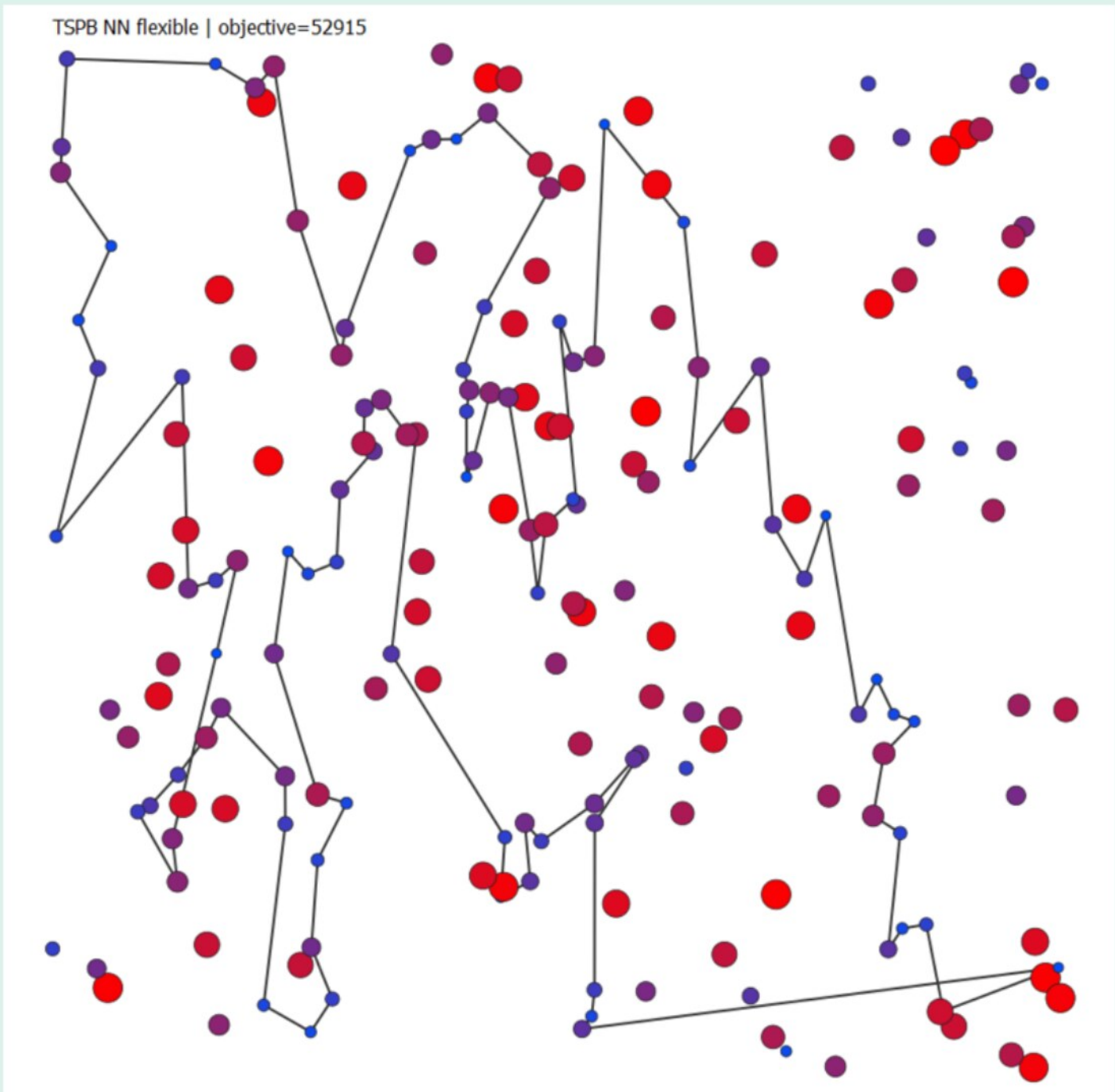
objective 56007

### Best solution nodes

```
89 57 76 54 127 129 90 153 20 14 2 110
118 56 114 196 10 148 143 78 192 162
104 185 184 3 151 69 172 133 120 96
121 198 48 72 46 86 117 160 136 8 21
154 140 47 170 125 177 6 132 29 197
175 134 45 60 40 71 155 97 115 180 65
28 149 145 158 116 147 42 50 130 30
183 38 64 85 59 167 51 87 179 82 88
171 99 67 24 34 181 152 73 98 27 135
75 79 156 43 89
```

100 selected nodes; closed route

# 7. Large best visualizations - TSPB




BEST SOLUTION

## Nearest Neighbor - Any Position

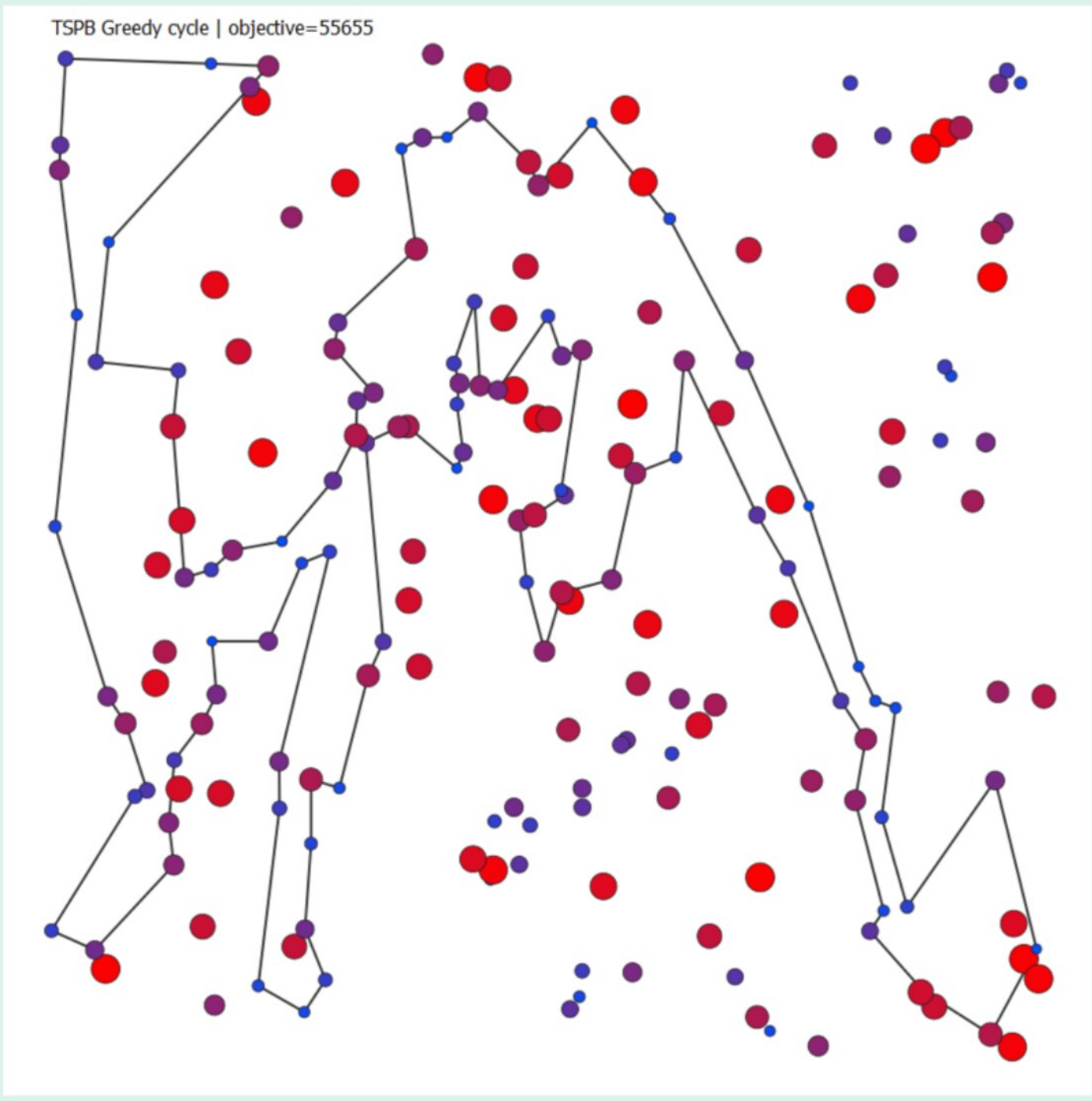
minimum 52915
maximum 66164
average 59480.28

**objective 52915**

### Best solution nodes

```
110 2 14 20 74 153 182 52 90 129 127
80 54 192 162 104 76 185 57 89 86 46
72 3 184 173 151 122 117 8 160 69 172
136 133 120 156 43 48 96 198 121 130
191 50 42 147 116 37 146 13 149 28 145
65 158 177 125 26 170 197 36 132 29 6
180 115 155 71 175 134 45 157 60 40
141 97 154 140 47 38 21 176 30 183 189
168 64 59 85 51 73 167 152 34 24 181
179 87 82 110
```

**100 selected nodes; closed route**




BEST SOLUTION

## Greedy Cycle

minimum 55655
maximum 63734
average 59649.91

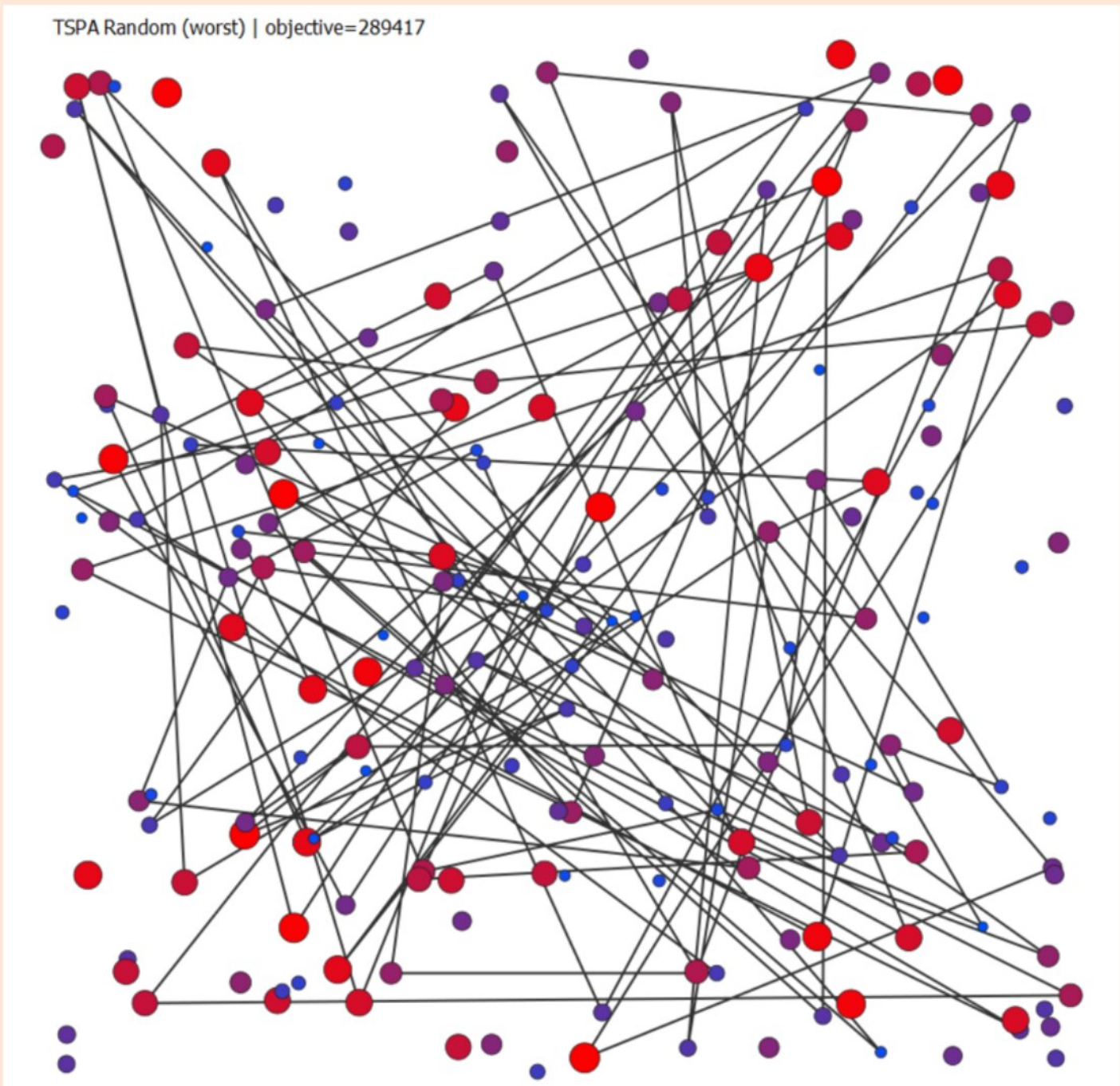
**objective 55655**

### Best solution nodes

```
174 36 197 6 180 115 170 97 140 47 155
71 175 134 45 157 60 141 40 33 64 21
189 168 69 160 172 136 133 120 8 117
72 46 86 184 3 173 122 151 98 111 27
135 76 185 162 192 80 52 182 20 74 2
93 110 118 14 153 90 129 127 54 104 57
89 156 43 48 96 198 121 63 130 191 183
30 176 38 154 26 125 177 138 158 145
149 147 42 116 37 146 13 28 65 178 103
29 132 1 174
```

**100 selected nodes; closed route**

# 7. Large worst visualizations - TSPA




WORST SOLUTION

## Random

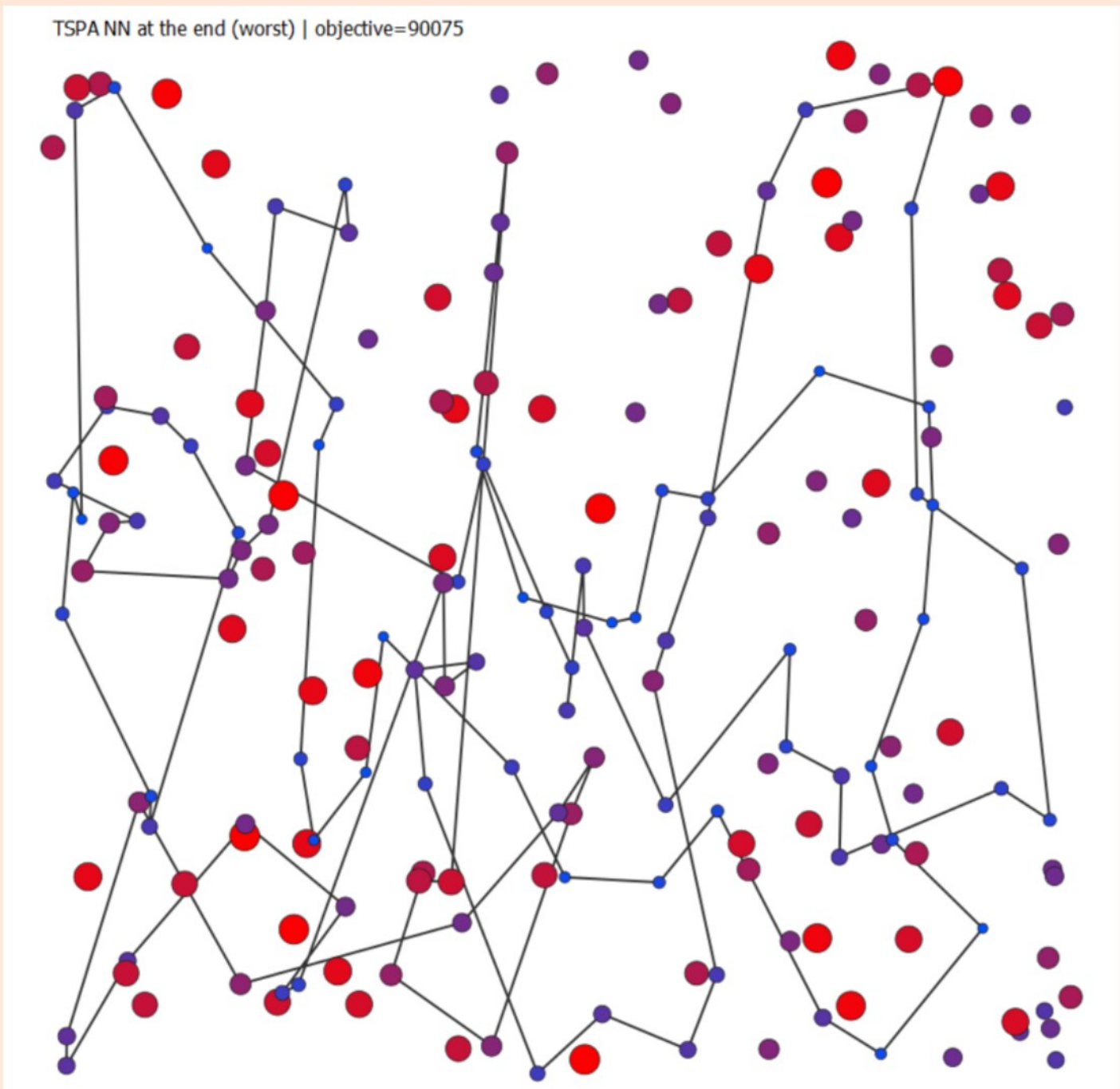
minimum 233130
maximum 289417
average 260294.40

objective 289417

### Worst solution nodes

109 161 155 165 12 148 144 51 160 95
65 139 167 189 24 61 14 195 182 90 184
23 123 172 36 120 108 199 198 186 192
194 187 140 125 145 176 142 130 43 113
68 0 154 54 143 96 103 185 151 7 27
134 107 115 81 162 57 31 13 75 66 133
6 118 146 8 128 152 183 67 127 173 116
158 10 156 99 38 163 110 26 28 136 49
88 119 46 181 137 20 55 40 82 177 166
17 175 124 149 109

100 selected nodes; closed route




WORST SOLUTION

## Nearest Neighbor - End

minimum 84362
maximum 90075
average 87476.63

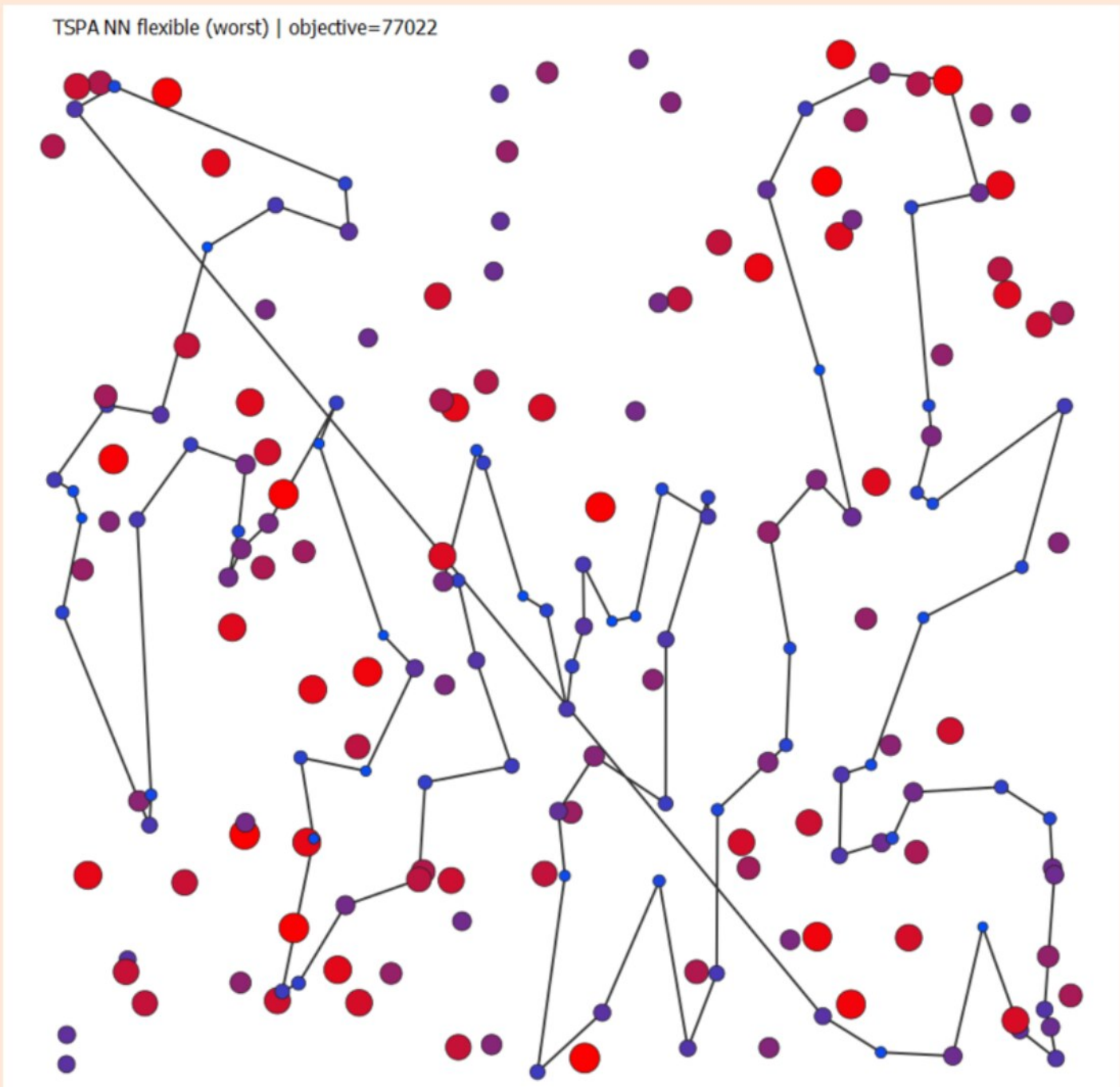
objective 90075

### Worst solution nodes

5 51 67 177 36 33 9 164 1 50 93 106 89
107 57 182 134 58 135 45 180 141 187
30 132 76 80 48 31 63 149 168 97 127
123 152 108 60 116 49 11 119 160 83 90
157 71 91 38 100 35 16 144 53 140 52
179 84 174 176 150 151 130 159 47 3
178 29 136 186 54 131 99 158 192 181
114 87 28 185 102 156 2 171 189 161 69
104 129 95 92 98 73 41 61 25 10 27 96
70 5

100 selected nodes; closed route

# 7. Large worst visualizations - TSPA




WORST SOLUTION

## Nearest Neighbor - Any Position

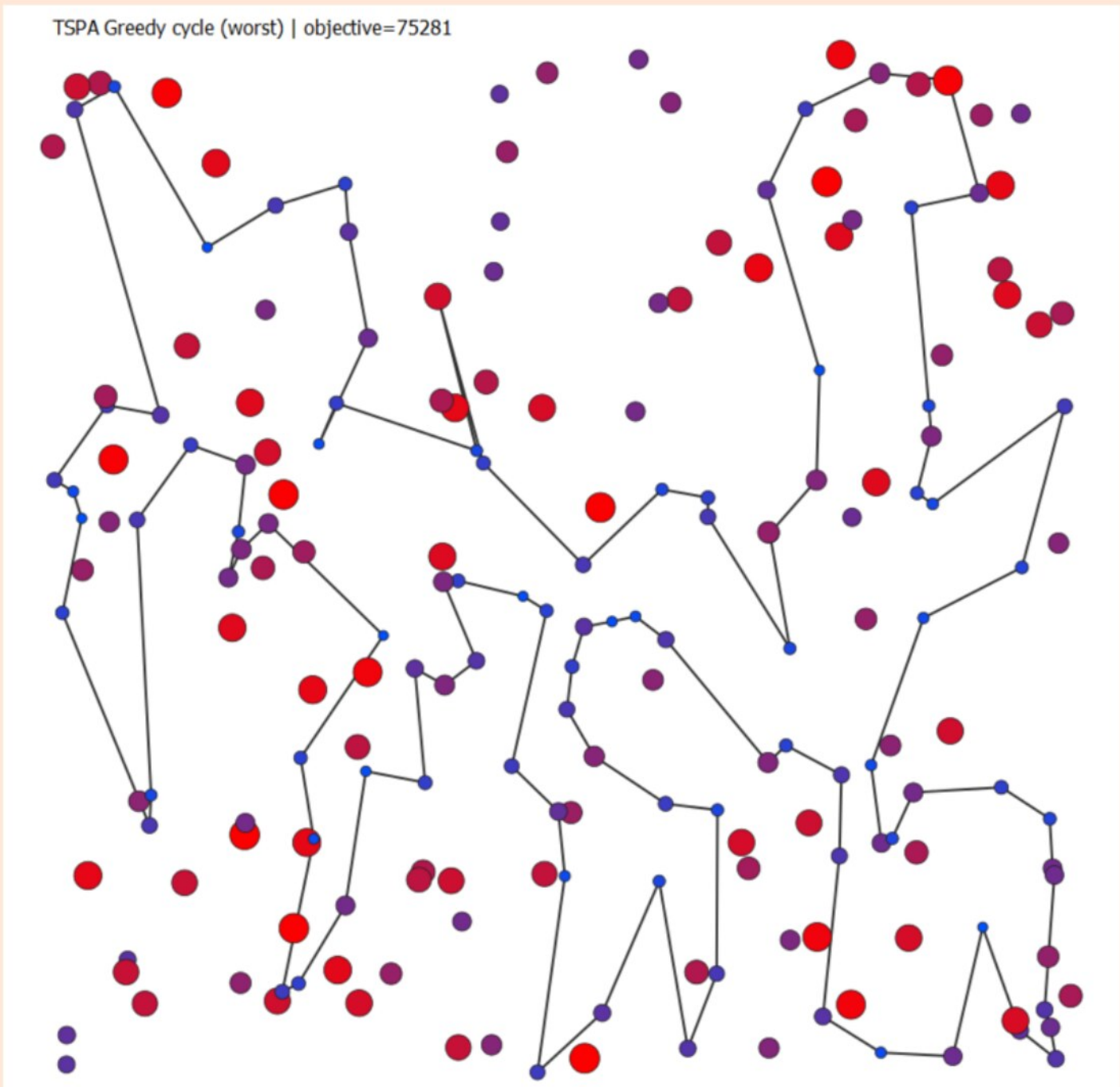
minimum 72328
maximum 77022
average 74346.97

objective 77022

### Worst solution nodes

```
31 48 157 71 91 80 108 60 116 149 63
168 95 127 97 49 152 100 123 160 83 90
76 132 180 28 141 30 187 171 2 189 96
87 45 185 35 156 51 16 67 144 140 53
179 52 177 36 33 54 9 131 84 41 73 135
114 181 58 192 158 134 72 176 174 8
142 85 164 186 136 163 29 101 178 1
169 3 50 147 47 93 106 150 151 42 89
113 130 159 146 170 153 193 78 88 107
64 57 182 31
```

100 selected nodes; closed route




WORST SOLUTION

## Greedy Cycle

minimum 72590
maximum 75281
average 73793.41

objective 75281

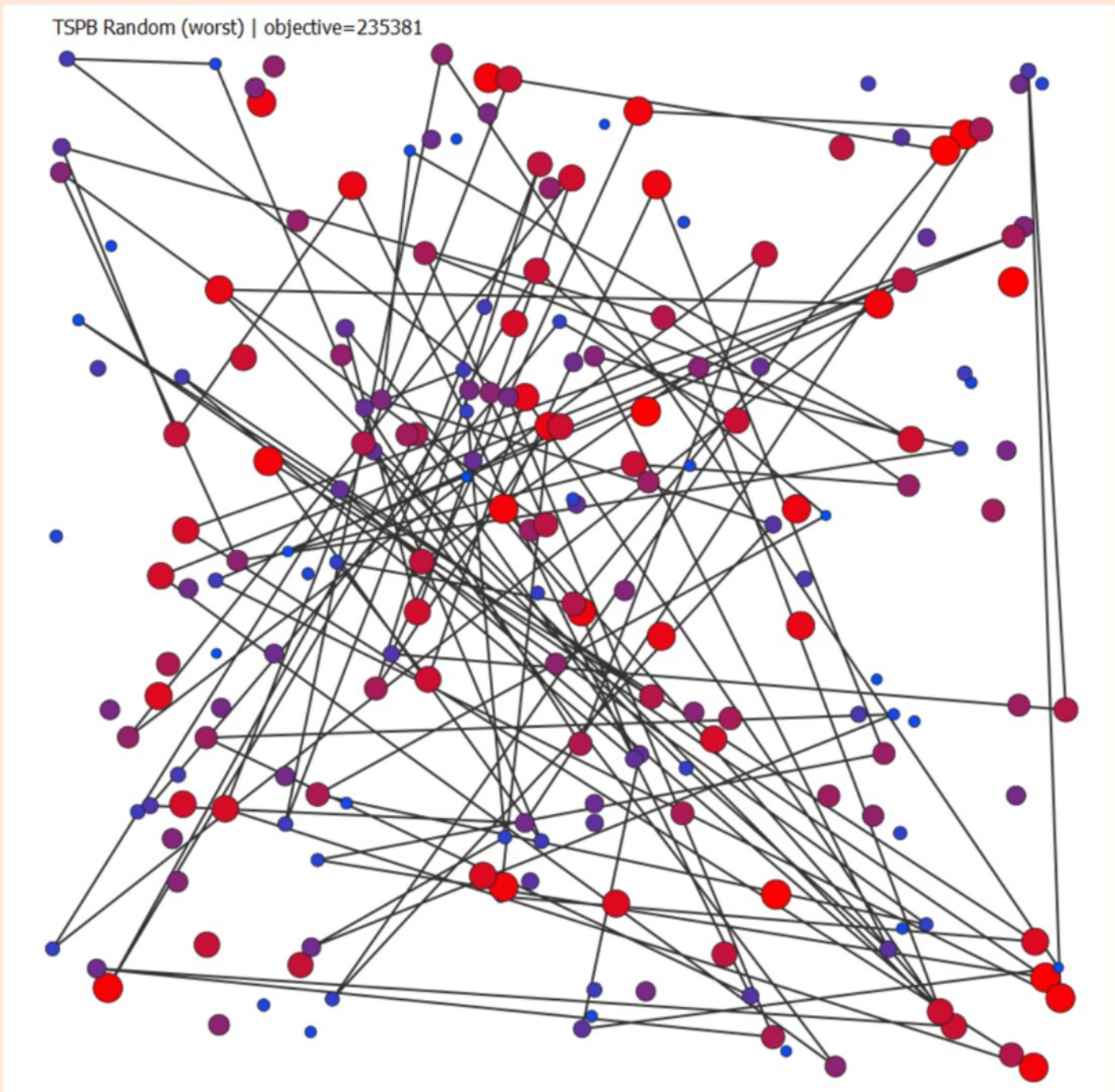
### Worst solution nodes

```
18 16 52 33 9 54 174 8 142 164 186 136
163 29 101 178 1 169 3 50 147 47 93
106 42 89 113 130 159 146 170 153 193
78 88 107 64 57 182 151 150 176 72 131
36 177 179 53 140 41 84 134 158 192 58
181 114 135 73 45 144 67 35 156 185
102 28 87 141 189 2 171 187 30 180 90
83 160 123 100 152 49 97 127 168 63
149 116 60 108 31 48 80 91 157 71 4
132 76 51 18
```

100 selected nodes; closed route

# 7. Large worst visualizations - TSPB






WORST SOLUTION

## Random

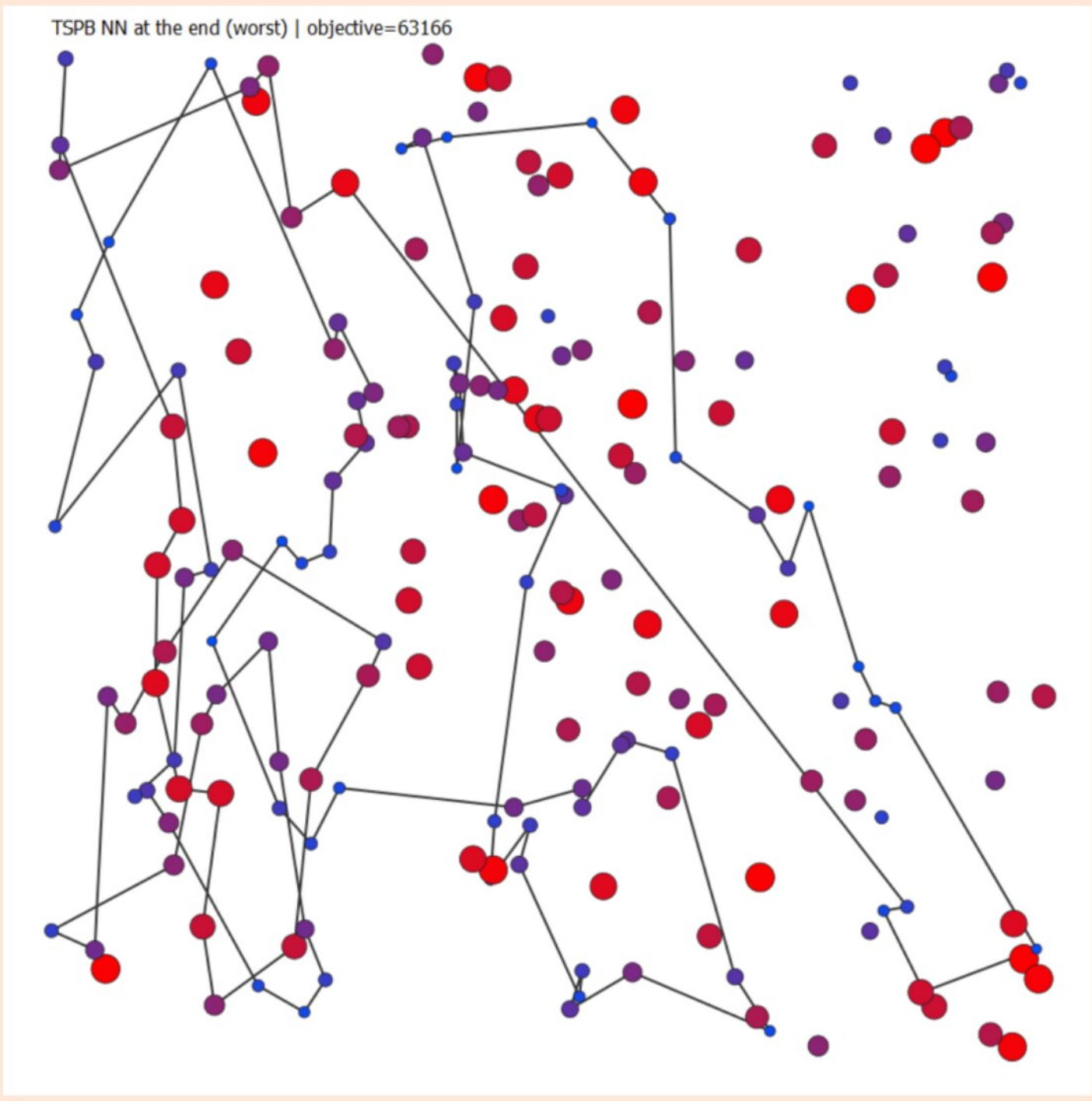
minimum 184618
maximum 235381
average 206384.53

objective 235381

### Worst solution nodes

26 13 173 63 41 121 183 162 45 144 72
29 73 169 44 64 131 94 112 125 12 123
9 174 39 117 37 116 167 161 11 158 34
22 130 4 32 103 77 25 30 133 59 157
129 180 99 195 138 146 79 78 75 154 18
1 38 2 128 83 52 60 100 28 68 85 109
33 43 71 58 110 82 24 168 108 166 54
141 14 124 185 188 95 49 47 111 199
164 113 23 55 139 53 17 176 67 74 86
10 26

100 selected nodes; closed route




WORST SOLUTION

## Nearest Neighbor - End

minimum 56007
maximum 63166
average 59397.34

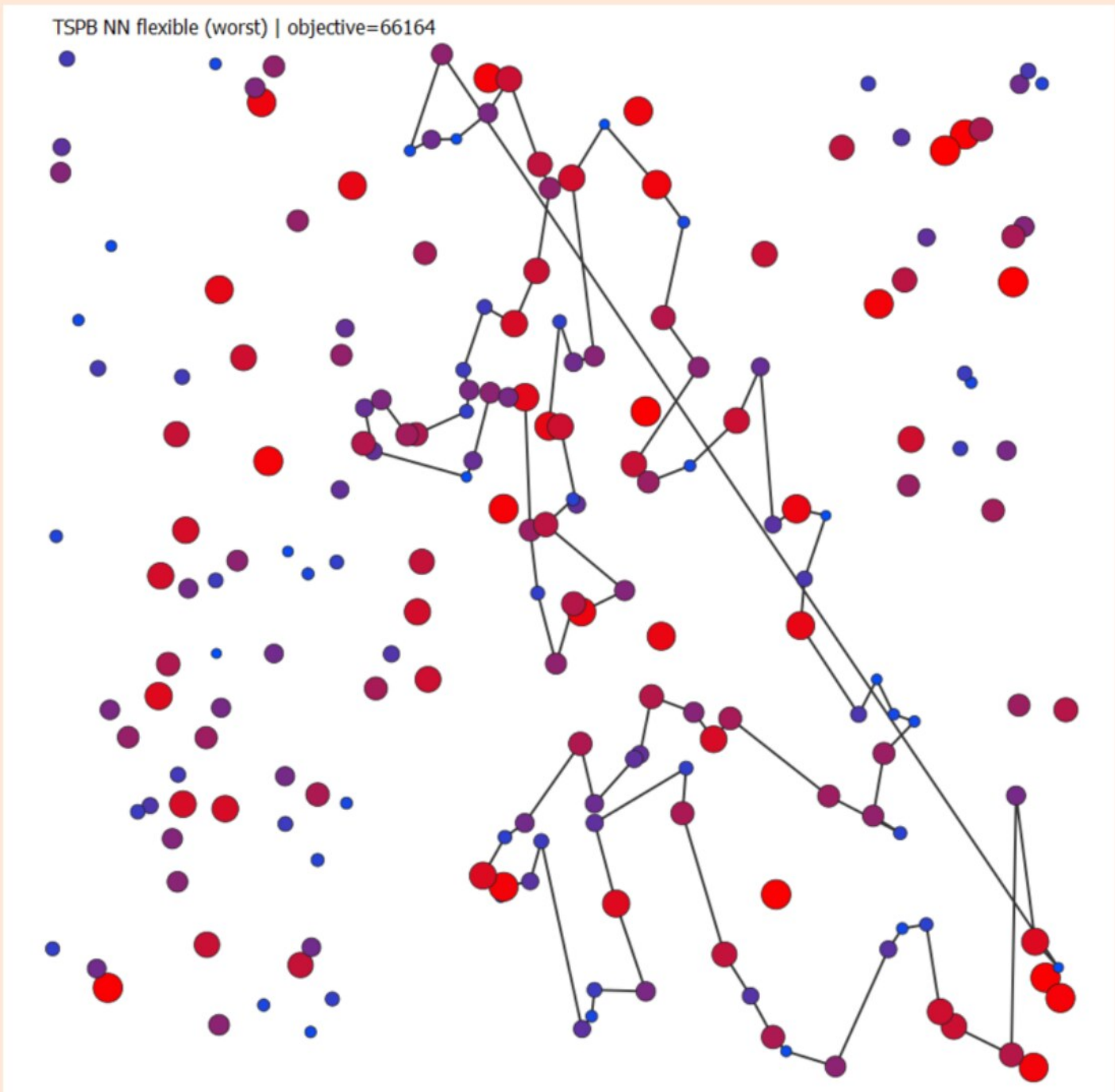
objective 63166

### Worst solution nodes

14 20 2 110 90 129 127 54 192 162 76
57 89 96 121 198 120 69 172 133 136
160 184 3 151 59 85 167 51 87 179 82
88 171 99 67 24 34 181 152 73 40 60 71
170 154 140 47 38 21 30 183 130 191
116 149 28 145 65 158 125 177 6 132 29
197 175 134 45 157 155 97 115 180 36 1
174 178 103 19 26 64 33 141 142 62 91
113 5 102 53 55 138 146 37 13 147 42
50 195 14

100 selected nodes; closed route

# 7. Large worst visualizations - TSPB




WORST SOLUTION

## Nearest Neighbor - Any Position

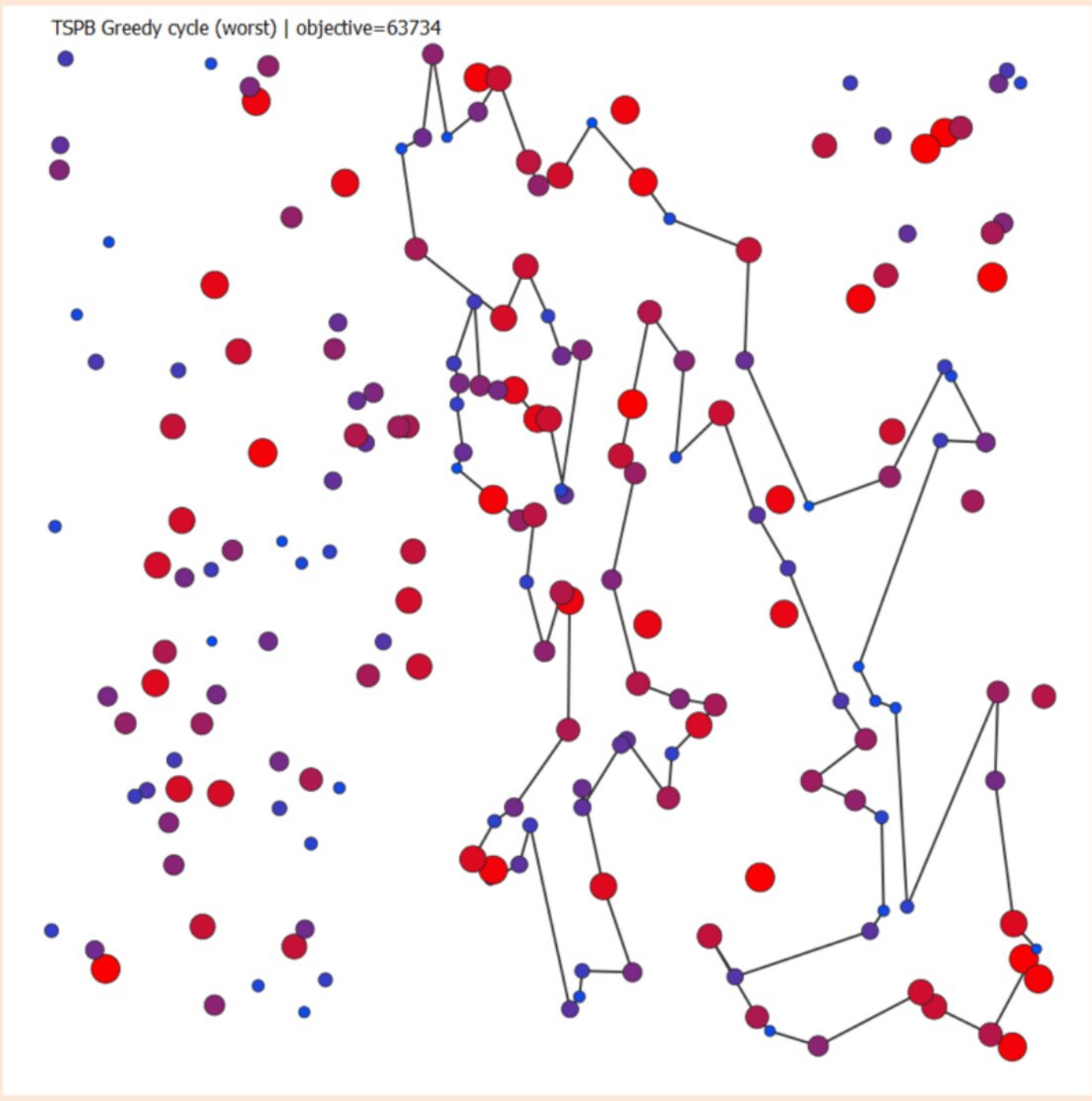
minimum 52915
maximum 66164
average 59480.28

**objective 66164**

### Worst solution nodes

110 68 118 93 9 2 188 14 20 74 17 171
39 99 163 31 67 181 126 88 179 87 82
167 51 101 85 107 59 73 25 152 34 24
11 66 119 84 150 153 182 52 90 129 127
80 106 192 54 159 162 104 144 76 135
75 185 79 57 83 89 32 86 46 72 109 137
3 184 173 27 22 111 98 151 122 95 117
8 160 69 21 30 183 189 168 172 136 133
120 92 166 156 43 164 48 96 198 121 58
110

**100 selected nodes; closed route**




WORST SOLUTION

## Greedy Cycle

minimum 55655
maximum 63734
average 59649.91

**objective 63734**

### Worst solution nodes

163 39 171 17 188 2 9 93 110 68 118 56
14 90 129 127 10 148 114 196 78 54 104
77 57 83 89 32 156 43 164 48 96 58 198
121 63 92 166 72 46 86 184 3 137 109
95 117 8 120 133 136 172 160 69 122
173 151 98 111 22 25 73 59 107 85 51
167 82 87 179 88 126 152 181 34 24 31
67 119 84 66 11 27 135 75 79 185 76
144 162 192 80 52 150 182 153 20 74 99
163

**100 selected nodes; closed route**